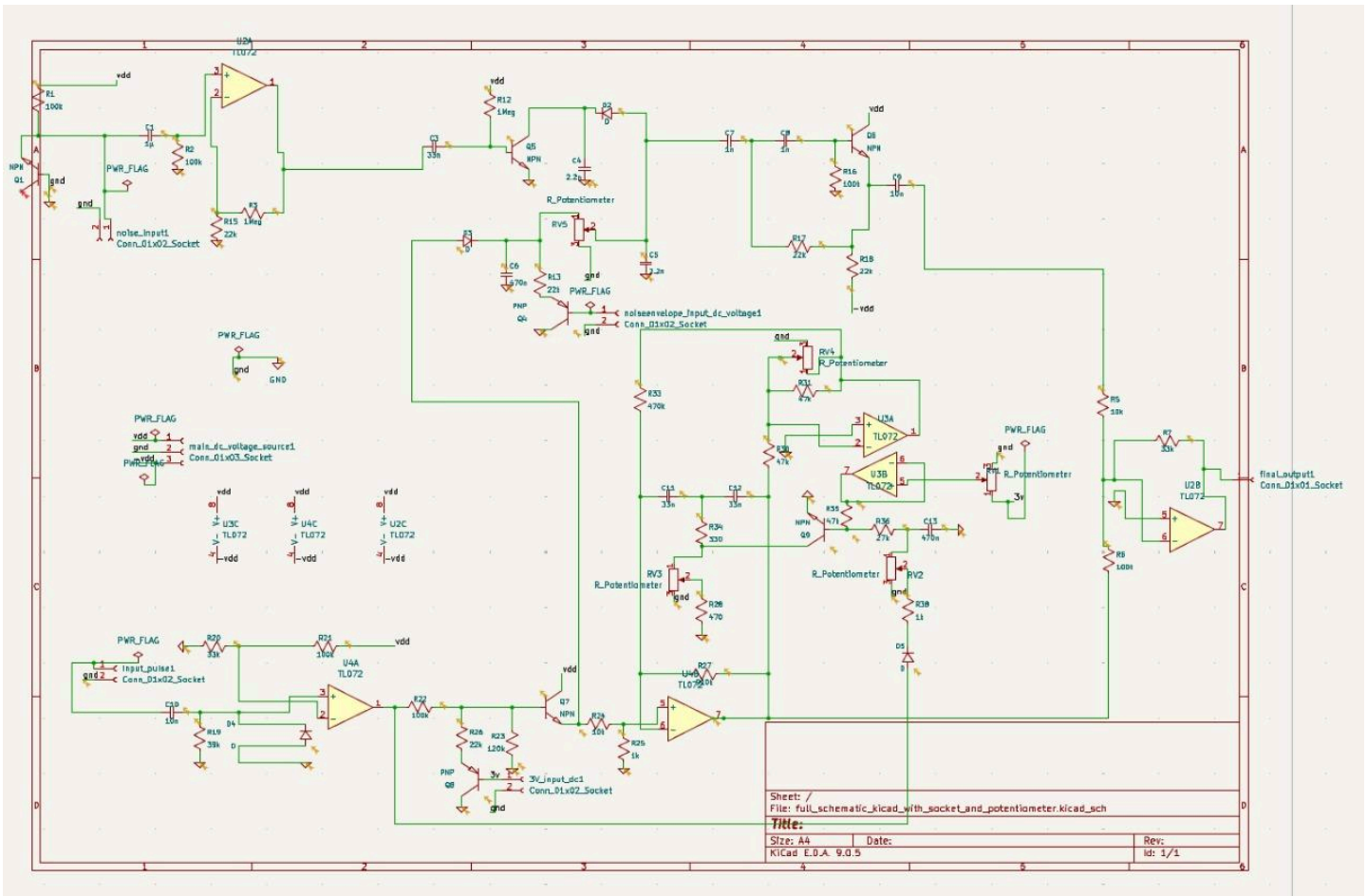


EE 322: AMSC
Sem 1 25-26
Team Behzad

Team Members:
Yash Sonone 23110318
Sumedh Wankhede 23110364
Ayush Umare 23110346

1. Schematic diagram:



2. Footprint Assignment Table:

Footprint Filters:   		
Footprint Libraries	Symbol : Footprint Assignments	Filtered Footprints
Audio_Module	1 3V_input_dc1 Conn_01x02_Socket : Connector_PinSocket 1.00mm:PinSocket_1x02_P1.00mm_Verical	1 Audio_Module:Reverb_BTDR-1H
Battery	2 C1 - 1µ : Capacitor_THT:CF_Radial_D5.0mm_P2.00mm	2 Audio_Module:Reverb_BTDR-1V
Button_Switch_Key	3 C3 - 33n : Capacitor_THT:C_Root_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	3 Battery:BatteryClip_Keystone_54_D16-19mm
Button_Switch_SMT	4 C4 - 2.2n : Capacitor_THT:C_Ract_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	4 Battery:BatteryHolder_Bulgin_8X0034_1xAAA
Button_Switch_THT	5 C5 - 2.2n : Capacitor_THT:C_Ract_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	5 Battery:BatteryHolder_Bulgin_8X0036_1xIC
Buzzer_Buzzer	6 C6 - 470n : Capacitor_THT:C_Root_L7.2mm_W4.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	6 Battery:BatteryHolder_Bulgin_8X0123_1xCR123
Calibration_Scale	7 C7 - 1n : Capacitor_THT:C_Root_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	7 Battery:BatteryHolder_ComfortableElectronic_CHZ
Capacitor_SMD	8 C8 - 1n : Capacitor_THT:C_Ract_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	8 Battery:BatteryHolder_Eagle_12H611-GR
Capacitor_Tantal	9 C9 - 10n : Capacitor_THT:C_Ract_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	9 Battery:BatteryHolder_Keystone_103_1x20mm
Capacitor_THT	10 C10 - 10n : Capacitor_THT:C_Ract_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	10 Battery:BatteryHolder_Keystone_104_1x23mm
Connector	11 C11 - 33n : Capacitor_THT:C_Root_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	11 Battery:BatteryHolder_Keystone_105_1x2430
Connector_AMAGS	12 C12 - 33n : Capacitor_THT:C_Root_L7.2mm_W2.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	12 Battery:BatteryHolder_Keystone_106_1x20mm
Connector_Amphenc	13 C13 - 470n : Capacitor_THT:C_Ract_L7.2mm_W4.5mm_P5.00mm_FK32_FKP2_MK32_MKP2	13 Battery:BatteryHolder_Keystone_107_1x23mm
Connector_Audio	14 D2 D : Diode_THT:D_DO-35_SOD27_P10.16mm_Horizontal	14 Battery:BatteryHolder_Keystone_500
Connector_Barrel3	15 D3 D : Diode_THT:D_DO-35_SOD27_P10.16mm_Horizontal	15 Battery:BatteryHolder_Keystone_590
Connector_Card	16 D4 D : Diode_THT:D_DO-35_SOD27_P10.16mm_Horizontal	16 Battery:BatteryHolder_Keystone_1042_1x18650
Connector_Coaxial	17 D5 D : Diode_THT:D_DO-35_SOD27_P10.16mm_Horizontal	17 Battery:BatteryHolder_Keystone_1057_1x2032
Connector_DIN	18 final_output1 - Conn_01x01_Socket : Connector_PinHeader 2.54mm:PinHeader_1x01_P2.54mm_Verical	18 Battery:BatteryHolder_Keystone_1058_1x2032
Connector_Dsub	19 input_pulse1 - Conn_01x02_Socket : Connector_PinHeader 2.54mm:PinHeader_1x02_P2.54mm_Verical	19 Battery:BatteryHolder_Keystone_1060_1x2032
Connector_FFC-FFC	20 main_dc_voltage_source1 - Conn_01x03_Socket : Connector_PinSocket 1.00mm:PinSocket_1x03_P1.00mm_Verical	20 Battery:BatteryHolder_Keystone_2460_1xAA
Connector_Harting	21 noiseEnvelope_input_dc_voltage1 - Conn_01x02_Socket : Connector_PinHeader 2.54mm:PinHeader_1x02_P2.54mm_Verical	21 Battery:BatteryHolder_Keystone_2462_2xAA
Connector_Harwin	22 noise_input1 - Conn_01x02_Socket : Connector_PinHeader 2.54mm:PinHeader_1x02_P2.54mm_Verical	22 Battery:BatteryHolder_Keystone_2466_1xAAA
Connector_Hirose	23 Q1 NPN : Package_TO_SOT_THT:TO-92_Inline	23 Battery:BatteryHolder_Keystone_2468_2xAAA
Connector_Hirose	24 Q4 - ENP : Package_TO_SOT_THT:TO-92_Inline	24 Battery:BatteryHolder_Keystone_2479_1xAAA
Connector_Hirose	25 Q5 - NPN : Package_TO_SOT_THT:TO-92_Inline	25 Battery:BatteryHolder_Keystone_2993
Connector_IUC	26 Q6 - NPN : Package_TO_SOT_THT:TO-92_Inline	26 Battery:BatteryHolder_Keystone_2998_1x6.8mm
Connector_JAK	27 Q7 - NPN : Package_TO_SOT_THT:TO-92_Inline	27 Battery:BatteryHolder_Keystone_3000_1x12mm
Connector_JAE_WP7	28 Q8 - ENP : Package_TO_SOT_THT:TO-92_Inline	28 Battery:BatteryHolder_Keystone_3001_1x12mm
Connector_JST	29 Q9 - NPN : Package_TO_SOT_THT:TO-92_Inline	29 Battery:BatteryHolder_Keystone_3002_1x2032
Connector_Molex	30 R1 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	30 Battery:BatteryHolder_Keystone_3008_1x2450
Connector_Molex_4	31 R2 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	31 Battery:BatteryHolder_Keystone_3009_1x2450
Connector_PCBBdge	32 R3 1Meg : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	32 Battery:BatteryHolder_Keystone_3034_1x20mm
Connector_Phoenix	33 R5 - 10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	33 Battery:BatteryHolder_LINK_BAT-HLD-012-SMT
Connector_Phoenix	34 R6 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	34 Battery:BatteryHolder_MPD_B5AVPC_1xPP3
Connector_Phoenix	35 R7 - 33k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	35 Battery:BatteryHolder_MPD_BC2AAPC_2xAA
Connector_Phoenix	36 R12 1Meg : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	36 Battery:BatteryHolder_MPD_BC12AAPC_2xAA
Connector_Phoenix	37 R13 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	37 Battery:BatteryHolder_MPD_BC2003_1x2032
Connector_Phoenix	38 R15 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	

Footprint Filters:   		
Footprint Libraries	Symbol : Footprint Assignments	Filtered Footprints
Audio_Module	30 R1 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	1 Audio_Module:Reverb_BTDR-1H
Battery	31 R2 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	2 Audio_Module:Reverb_BTDR-1V
Button_Switch_Key	32 R3 - 1Meg : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	3 Battery:BatteryClip_Keystone_54_D16-19mm
Button_Switch_SMT	33 R5 - 10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	4 Battery:BatteryHolder_Bulgin_8X0034_1xAAA
Button_Switch_THT	34 R6 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	5 Battery:BatteryHolder_Bulgin_8X0036_1xIC
Buzzer_Buzzer	35 R7 - 33k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	6 Battery:BatteryHolder_Bulgin_8X0123_1xCR123
Calibration_Scale	36 R12 - 1Meg : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	7 Battery:BatteryHolder_ComfortableElectronic_CHZ
Capacitor_SMD	37 R13 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	8 Battery:BatteryHolder_Eagle_12H611-GR
Capacitor_Tantal	38 R15 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	9 Battery:BatteryHolder_Keystone_103_1x20mm
Capacitor_THT	39 R16 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	10 Battery:BatteryHolder_Keystone_104_1x23mm
Connector	40 R17 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	11 Battery:BatteryHolder_Keystone_105_1x2430
Connector_AMAGS	41 R18 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	12 Battery:BatteryHolder_Keystone_106_1x20mm
Connector_Amphenc	42 R19 - 33k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	13 Battery:BatteryHolder_Keystone_107_1x23mm
Connector_Audio	43 R20 - 33k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	14 Battery:BatteryHolder_Keystone_500
Connector_Barrel3	44 R21 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	15 Battery:BatteryHolder_Keystone_590
Connector_Card	45 R22 - 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	16 Battery:BatteryHolder_Keystone_1042_1x18650
Connector_Coaxial	46 R23 - 120k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	17 Battery:BatteryHolder_Keystone_1057_1x2032
Connector_DIN	47 R24 - 10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	18 Battery:BatteryHolder_Keystone_1058_1x2032
Connector_Dsub	48 R25 - 1k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	19 Battery:BatteryHolder_Keystone_1060_1x2032
Connector_FFC-FFC	49 R26 - 22k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	20 Battery:BatteryHolder_Keystone_2460_1xAA
Connector_Harting	50 R27 - 910k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	21 Battery:BatteryHolder_Keystone_2462_2xAA
Connector_Harwin	51 R28 - 470 : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	22 Battery:BatteryHolder_Keystone_2466_1xAAA
Connector_Hirose	52 R30 - 47k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	23 Battery:BatteryHolder_Keystone_2468_2xAAA
Connector_Hirose	53 R31 - 47k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	24 Battery:BatteryHolder_Keystone_2479_1xAAA
Connector_Hirose	54 R33 - 470k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	25 Battery:BatteryHolder_Keystone_2993
Connector_IUC	55 R34 - 330 : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	26 Battery:BatteryHolder_Keystone_2998_1x6.8mm
Connector_JAE_WP7	56 R35 - 47k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	27 Battery:BatteryHolder_Keystone_3000_1x12mm
Connector_JST	57 R36 - 27k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	28 Battery:BatteryHolder_Keystone_3001_1x12mm
Connector_Molex	58 R38 - 1k : Resistor_THT:R_Axial_DIN0207_L6.3mm_W2.5mm_P10.16mm_Horizontal	29 Battery:BatteryHolder_Keystone_3002_1x2032
Connector_Molex	59 RV1 R_Potentiometer : Potentiometer_THT:Potentiometer_Bourns_3386F_Verical	30 Battery:BatteryHolder_Keystone_3008_1x2450
Connector_Molex_4	60 RV2 R_Potentiometer : Potentiometer_THT:Potentiometer_Bourns_3386F_Verical	31 Battery:BatteryHolder_Keystone_3009_1x2450
Connector_PCBBdge	61 RV3 R_Potentiometer : Potentiometer_THT:Potentiometer_Bourns_3386F_Verical	32 Battery:BatteryHolder_Keystone_3034_1x20mm
Connector_Phoenix	62 RV4 R_Potentiometer : Potentiometer_THT:Potentiometer_Bourns_3386F_Verical	33 Battery:BatteryHolder_LINK_BAT-HLD-012-SMT
Connector_Phoenix	63 RV5 R_Potentiometer : Potentiometer_THT:Potentiometer_Bourns_3386F_Verical	34 Battery:BatteryHolder_MPD_B5AVPC_1xPP3
Connector_Phoenix	64 U2 TL072 : Package_DIP:DIP-8_W7.62mm	35 Battery:BatteryHolder_MPD_BC2AAPC_2xAA
Connector_Phoenix	65 U3 TL072 : Package_DIP:DIP-8_W7.62mm	36 Battery:BatteryHolder_MPD_BC12AAPC_2xAA
Connector_Phoenix	66 U4 TL072 : Package_DIP:DIP-8_W7.62mm	37 Battery:BatteryHolder_MPD_BC2003_1x2032

Filtered by Library: 15205 matching footprints

3. ERC report:

Electrical Rules Checker

Violations (89) Ignored Tests (4)

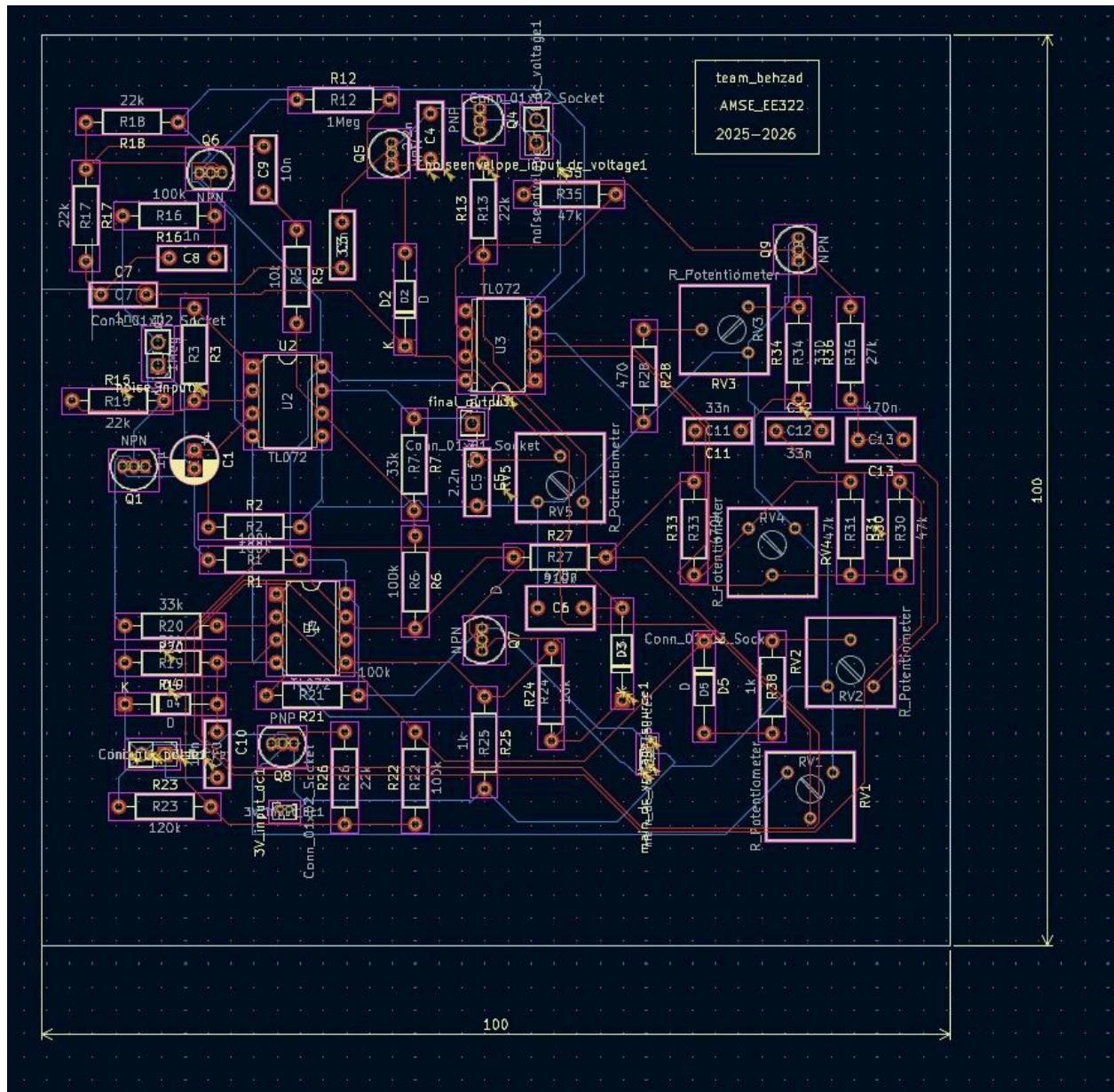
- ✓ **Error: Pin not connected**
Symbol Q1 Pin 1 [C, Passive, Line]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol R1 [res]
- ✓ **Warning: Power pin and label are similar (lower/upper case difference only)**
Label 'gnd'
Symbol #PWR01 Pin 1 [Power input, Line]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol Q1 [npn]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol #GND001 [GND]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol C1 [cap]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol R2 [res]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol #GND000 [GND]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol R15 [res]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol #GND008 [GND]
- ✓ **Warning: The current configuration does not include the symbol library 'ItsPice'**
Symbol R3 [res]
- ✓ **Warning: Both 0 and gnd are attached to the same items; 0 will be used in the netlist**
Symbol #GND000 Pin [Power input, Line]

Show: ☐ All ☒ Errors 1 ☒ Warnings 88 ☐ Exclusions

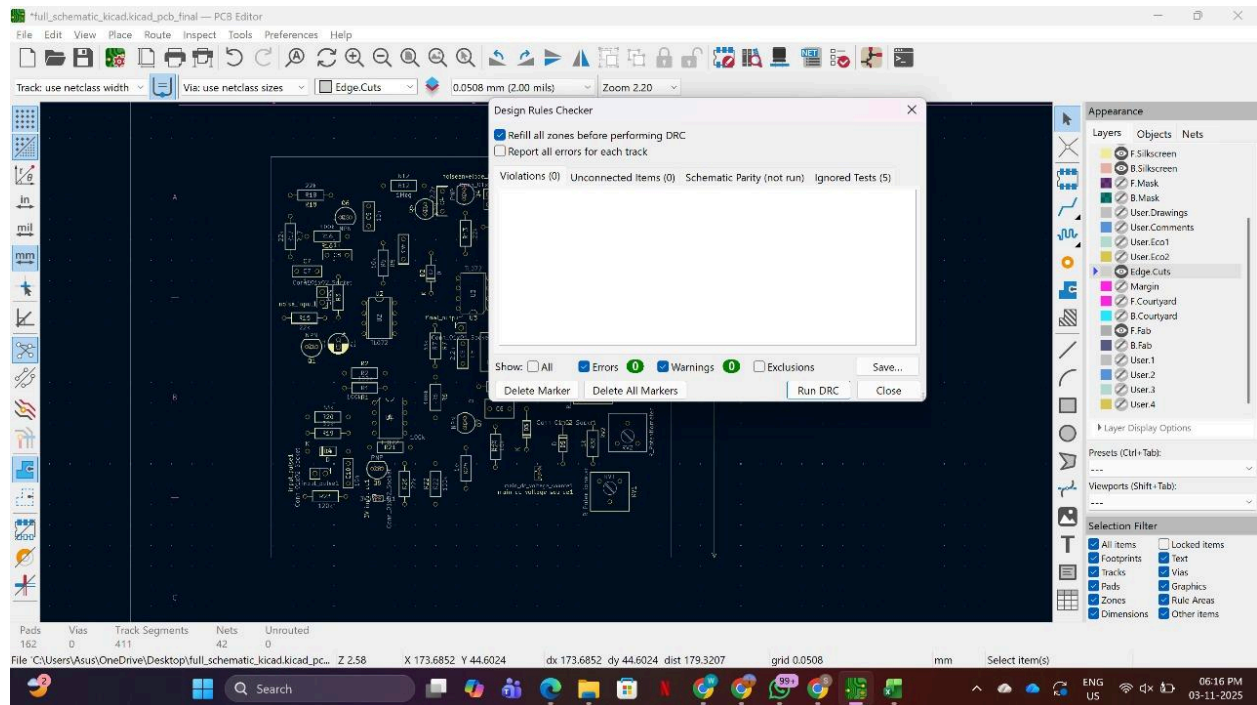
Delete Marker Delete All Markers

Save... Run ERC Close

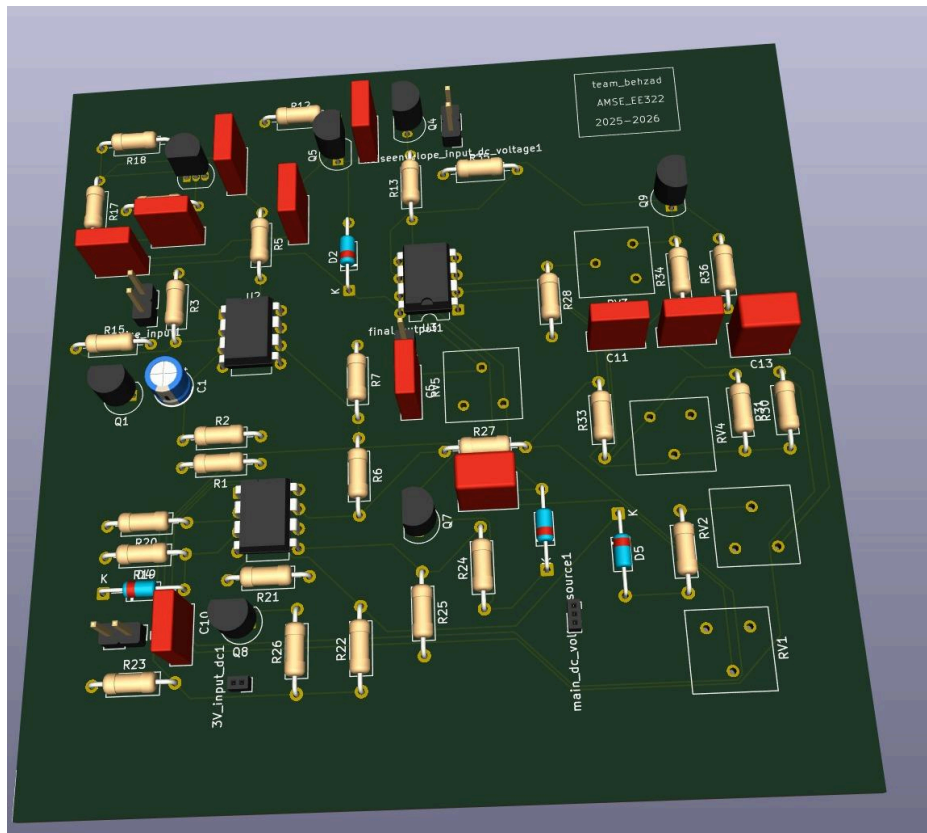
4. Layout:

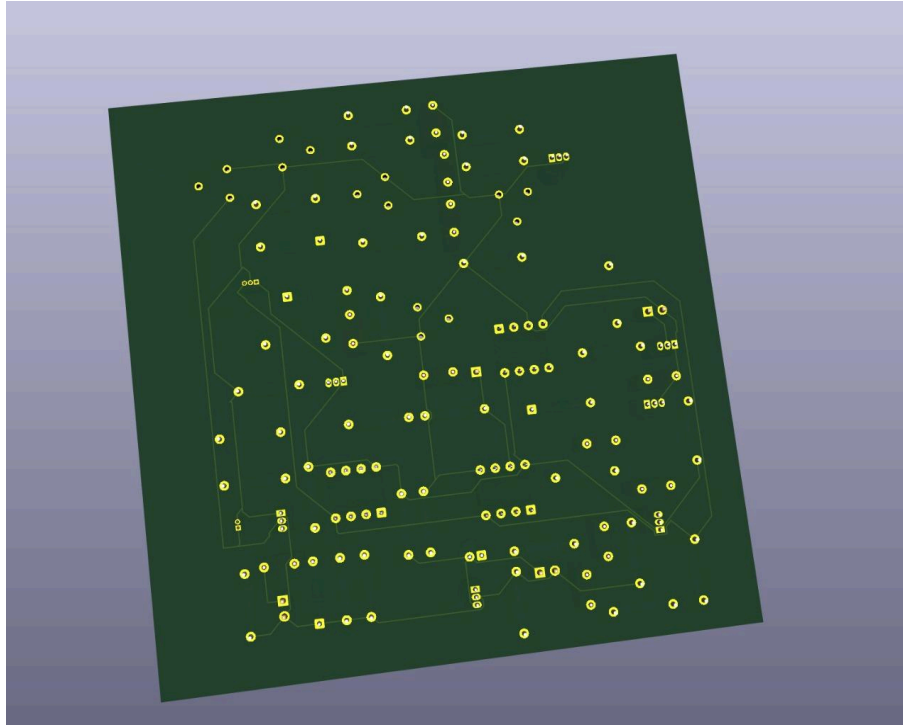


5. DRC report:



6. 3D rendering of the designed PCB:





7. Screenshot of the Gerber files:

 full_schematic_kicad-B_Cu.gbr		01-11-2025 07:58 PM	GBR File	23 KB
 full_schematic_kicad-B_Mask.gbr		01-11-2025 07:58 PM	GBR File	8 KB
 full_schematic_kicad-B_Paste.gbr		01-11-2025 07:58 PM	GBR File	1 KB
 full_schematic_kicad-B_Silkscreen.gbr		01-11-2025 07:58 PM	GBR File	1 KB
 full_schematic_kicad-drl.rpt		01-11-2025 07:57 PM	RPT File	1 KB
 full_schematic_kicad-Edge_Cuts.gbr		01-11-2025 07:58 PM	GBR File	1 KB
 full_schematic_kicad-F_Cu.gbr		01-11-2025 07:58 PM	GBR File	26 KB
 full_schematic_kicad-F_Mask.gbr		01-11-2025 07:58 PM	GBR File	8 KB
 full_schematic_kicad-F_Paste.gbr		01-11-2025 07:58 PM	GBR File	1 KB
 full_schematic_kicad-F_Silkscreen.gbr		01-11-2025 07:58 PM	GBR File	176 KB
 full_schematic_kicad-job.gbrjob		01-11-2025 07:58 PM	GBRJOB File	3 KB
 full_schematic_kicad-NPTH.drl		01-11-2025 07:58 PM	DRL File	1 KB
 full_schematic_kicad-PTH.drl		01-11-2025 07:58 PM	DRL File	4 KB
 report.txt		01-11-2025 07:53 PM	Text Document	1 KB

8. Gerber files uploaded on lionscircuit.com:

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Project Description

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Reset

Base Material

FH4

Layers

246810121416182022

Dimensions

100 x 100 mm

Quantity

5

Discrete Design

12345

Delivery Format

Single PCBPanel By CustomerPanel By LionCircuits

PCB Thickness (mm)

0.40.81.21.62.02.4

Mask Color

GreenWhiteRedBlueBlack

PCB Finish

HASL FinishLead Free HASLENIG

Copper Thickness

1 oz (35 um)2 oz (70 um)3 oz (105 um)

Charge Details

Make In India

Close

Price Estimate

Per Unit Cost ₹322.20 × 5

NRE Cost ₹0

Sub Total ₹1611

Build Time

☒ 5-6 Days☐ 4-5 Days

Shipping Method

☒ DTDG Standard 2-3 Working Days☐ DTDG Plus 1-2 Working Days

New Sub Total ₹1611

GST ₹290

Shipping Cost ₹0

Total Cost ₹1901

Save to Cart

10. LTspice Schematic and Simulation results:

